IV B. Tech I Semester

23AD41T1–GENERATIVE AI

Course Category:	Professional core	**Credits:**	3
Course Type:	Theory	**Lecture-Tutorial-Practical:**	3-0-0
Prerequisite:	• Knowledge in Generative AI	**Sessional Evaluation:** **Univ. Exam Evaluation:** **Total Marks:**	30 70 100
Course Objectives:	**Students undergoing this course are expected:** • Introduce the fundamentals of Generative AI, including its principles, architecture, and evolution. • Provide a deep understanding of Large Language Models (LLMs) and their application in natural language generation tasks. • Develop practical knowledge of Prompt Engineering, including prompt tuning, prompt design, and performance evaluation. • Explore applications of Generative AI across various domains including code generation, art synthesis, content generation, and interactive systems. • Equip students with ethical, social, and safety considerations when designing and deploying generative AI applications..		

Course Outcomes:	**Upon successful completion of the course, the students will be able to:**	
	CO1	Demonstrate a strong understanding of the architecture and functioning of Generative AI models, including transformers and LLMs.
	CO2	Be capable of applying prompt engineering techniques to steer model behavior for desired outputs across various tasks..
	CO3	Design and fine-tune generative models for applications such as text generation, image creation, music synthesis, and conversational AI...
	CO4	Analyze and evaluate the effectiveness of prompts and generated content, using relevant metrics and methodologies.
	CO5	Apply ethical principles to ensure responsible development and deployment of generative AI systems.
Course Content:	**UNIT-I** **Introduction to Generative AI(Cognitive Level: Understand, Remember):** Overview of AI and types of AI, What is Generative AI? Definitions and Concepts, Historical evolution of generative models, Types of generative models – GANs, VAEs, Autoregressive Models, Introduction to Transformers and LLMs, Applications and use cases of Generative AI, Challenges in Generative AI development, Introduction to text-to-image and image-to-text models **UNIT-II** **Fundamentals of Prompt Engineering**: Definition and significance of Prompt Engineering, Types of prompts: Zero-shot, One-shot, Few-shot, Techniques for effective prompt design, Prompt templates and chaining, Prompt tuning and parameter-efficient tuning, Evaluating prompt performance, Use of APIs for testing prompts (OpenAI, Cohere, Anthropic), Best practices and prompt libraries.	

	UNIT-III **Working with Large Language Models (LLMs):** Overview of pre-trained LLMs: GPT, BERT, LLaMA, Claude, PaLM, Architectures and tokenization strategies, Fine-tuning vs. in-context learning, LLM-powered tools (ChatGPT, GitHub Copilot, Bard, Claude), Role of attention mechanism and transformer layers, Tools for model experimentation (Hugging Face, LangChain), Performance metrics for LLMs, Case studies of model adaptation and deployment. **UNIT-IV** **Applications of Generative AI:** Text generation and summarization, Image and art generation (DALL·E, Midjourney, Stable Diffusion), Code generation and completion tools (Codex, Copilot), Music and video generation, Generative chatbots and customer service, Story generation and dialogue systems, Domain-specific applications (Legal, Healthcare, Education), Comparative study of generative models by task. **UNIT-V** **Ethics, Security & Responsible AI**: Bias and fairness in LLMs and generative systems, Explainability and transparency in generative AI, Copyright and originality issues, Adversarial use of generative models – deepfakes, misinformation, AI safety protocols and red-teaming, Regulatory and policy frameworks for generative AI, Responsible prompt crafting and moderation
Text Books & References Books:	**TEXTBOOKS:** 1. Deep Learning with Python by François Chollet, Manning Publications 2. Transformers for Natural Language Processing by Denis Rothman, Packt 3. Practical Generative AI by Amit Shukla, BPB Publications **REFERENCE BOOKS:** 1. Generative Deep Learning by David Foster, O'Reilly 2. Artificial Intelligence: A Guide for Thinking Humans by Melanie Mitchell 3. Machine Learning B.Teching by Andrew Ng (available online)
Online Learning Resources:	1. Generative AI with Large Language Models – Coursera (AWS & DeepLearning.AI) 2. Prompt Engineering for ChatGPT – Coursera 3. Generative AI Fundamentals – Google Cloud Training